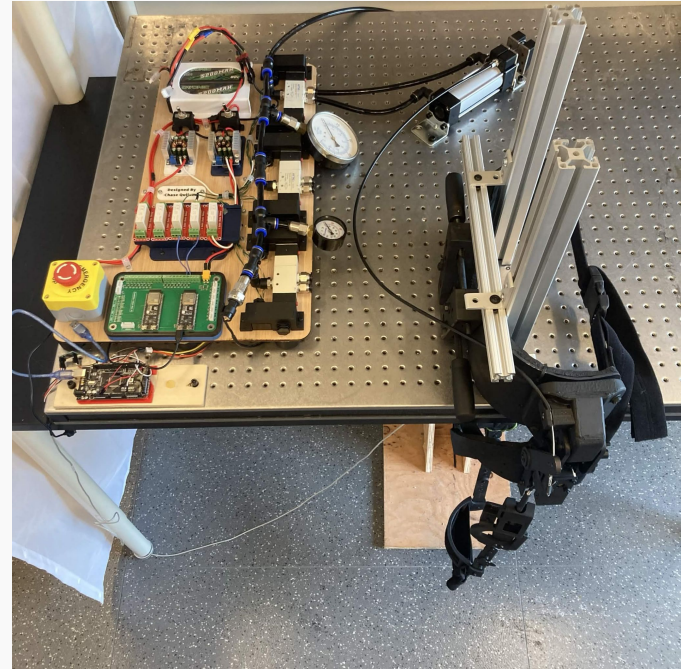


Final Review

Fall Prevention

Exoskeleton Clinic

Chase Quijano, Eric Carty, Preston Haddon



Introductions



Chase Quijano

Engineering
Entrepreneurship
(Mechanical Track)



Eric Carty

Mechanical Engineering



Preston Haddon

Electrical Engineering

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- Regulations Adherence
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Problem Statement

Slips and falls are the leading cause of injuries among the elderly.

The Solution:



Lightweight
Wearable
Support

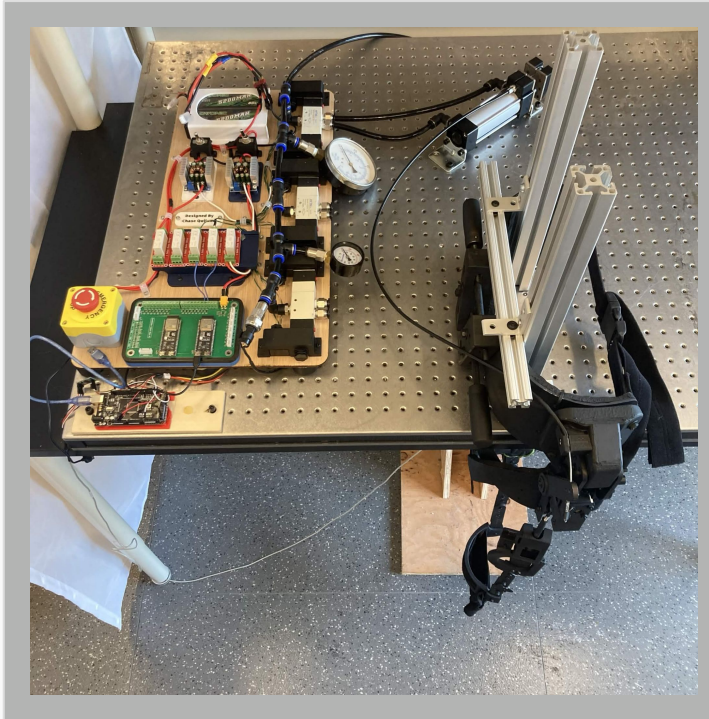


Realtime Gait
Detection



Pneumatic
Reactive Fall
Correction

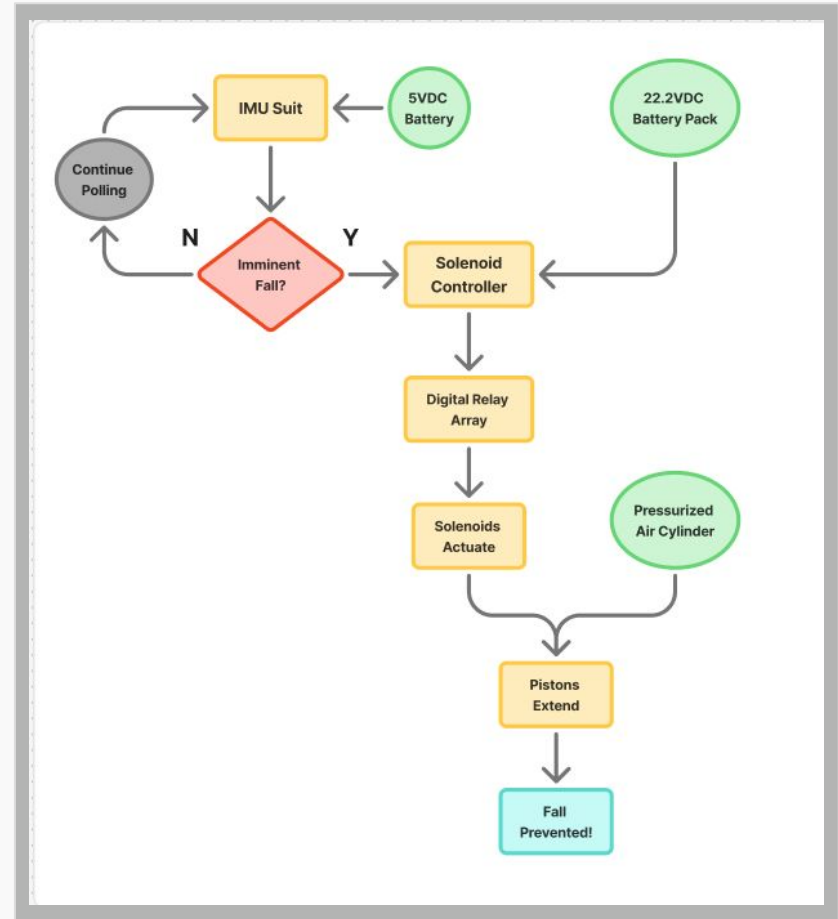
The Fall-Prevention Exoskeleton



Design Overview

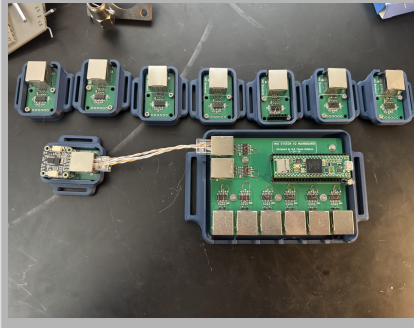
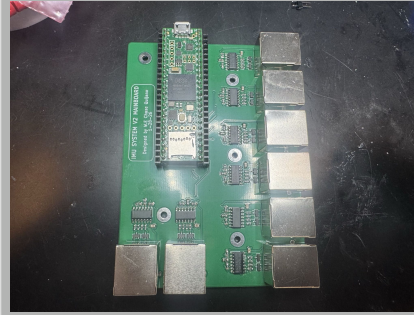
The Fall-Prevention Exoskeleton is comprised of several key components:

- Control System (relays, batteries, control board, solenoids)
- Actuators (pneumatic cylinders)
- Chassis
- IMU System
- Gait Detection and Correction



IMU System V2

- 400Hz sampling speed between 8 IMUs
- Upgraded communication protocol (UART over RS485)
- Custom Shielded Cat6 cabling with twisted pairs
- Utilizes full duplex transceivers
- Dedicated microcontroller for sending IMU data to primary computer (Raspberry Pi)



The IMU system is worn in conjunction with the exoskeleton and will provide real time gait detection and correction



IMU System DAQ

- IMU System is accessed via SSH where a logging script is ran
- Data is logged in quaternions and imported into OpenSim and MatLab for analysis
 - Quaternions are the unit of data that we use to track positioning in space



The IMU nodes are fastened to Thigh, Calf, Ankle, Lower back and upper back.

OLD LEG CHASSIS

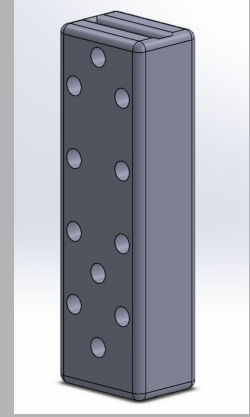
- Faults
 - Didn't transfer forces well
- Parts that needed improvement
 - Thigh links
 - Thigh brace



NEW LEG CHASSIS

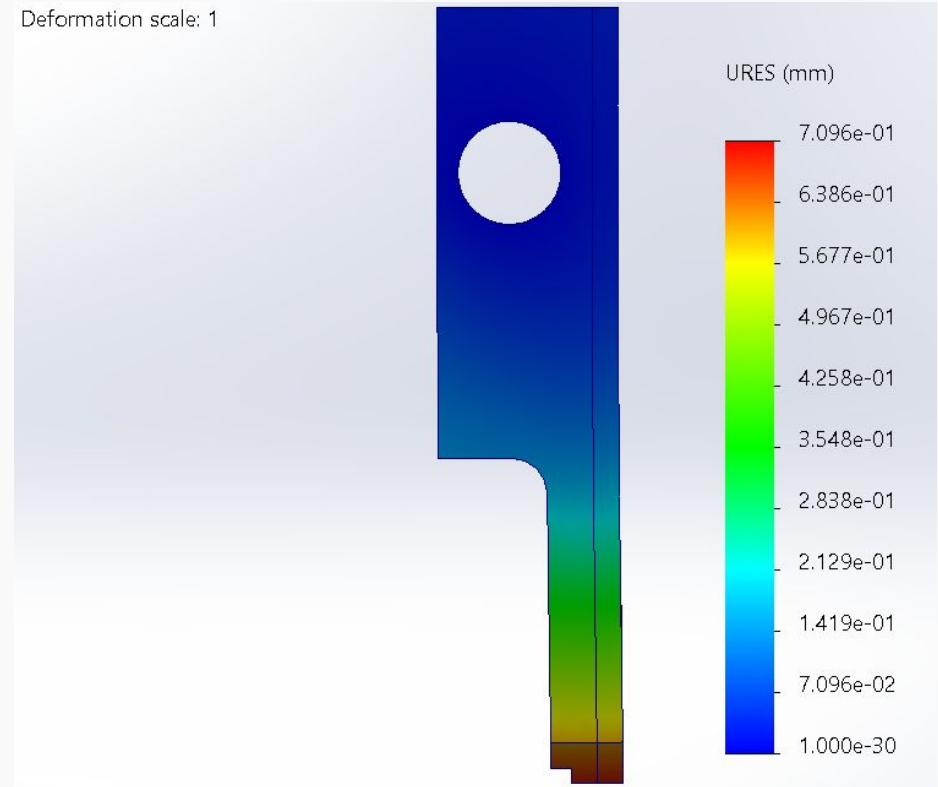
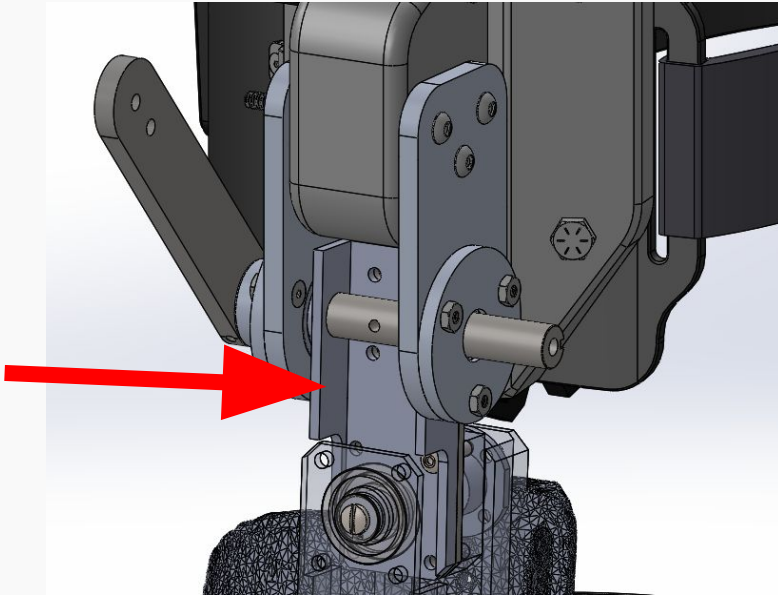
Parts improved:

- Thigh links
- Thigh brace
- Fancy paint job



Thigh Link Simulation (3000 N)

Deformation scale: 1

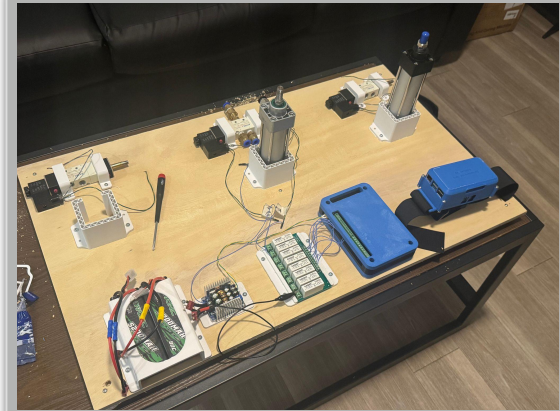


Test Configuration and Reliability

The most immediate and pressing concern during this semester appeared during our first iterations on the solenoid controller: **How do we ensure it is configured correctly every time?**

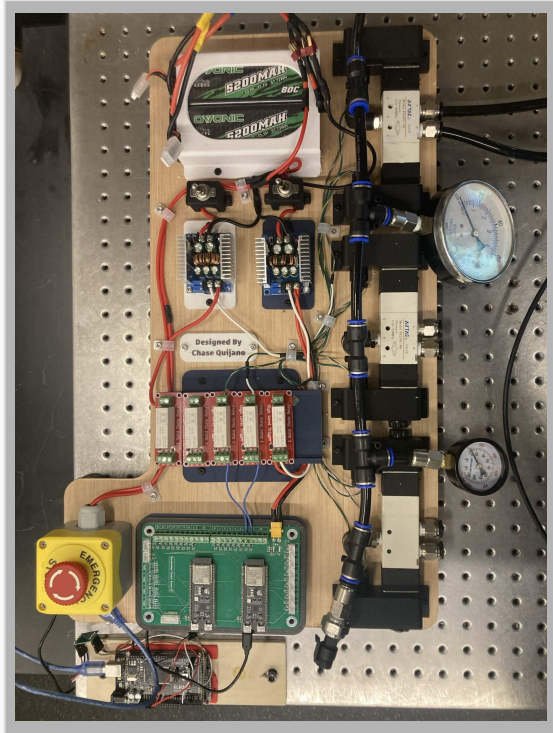
Setup Considerations:

- Component Polarity Consistency
- Emergency Shutoff
- Consistent Component Grounding
- Effective Pneumatic Routing
- Battery Charge and Configuration
- Multi-Stage Voltage Regulation
- Electrical Feedback Protection



Initial Test Bench Design
circa early December 2025

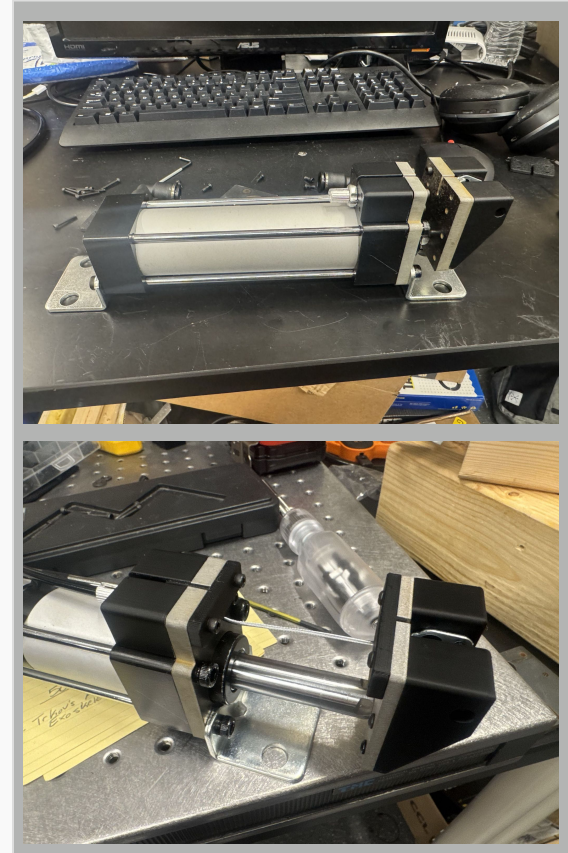
Refined Test Bench



- Emergency Shutoff Implementation
 - E-Stop button is readily available during emergency
- Dedicated Line Regulation Improvements
 - Switches for manual toggling of 15v and 5v lines
- Centralized Pneumatic Line
 - All solenoids are pressurized by a single pneumatic line for ease of testing
- Intuitive Layout & Mounting Solutions
 - Proper Cable Management
- Integrated Pressure Transducer
 - Assists with data acquisition by providing real-time pressure readings

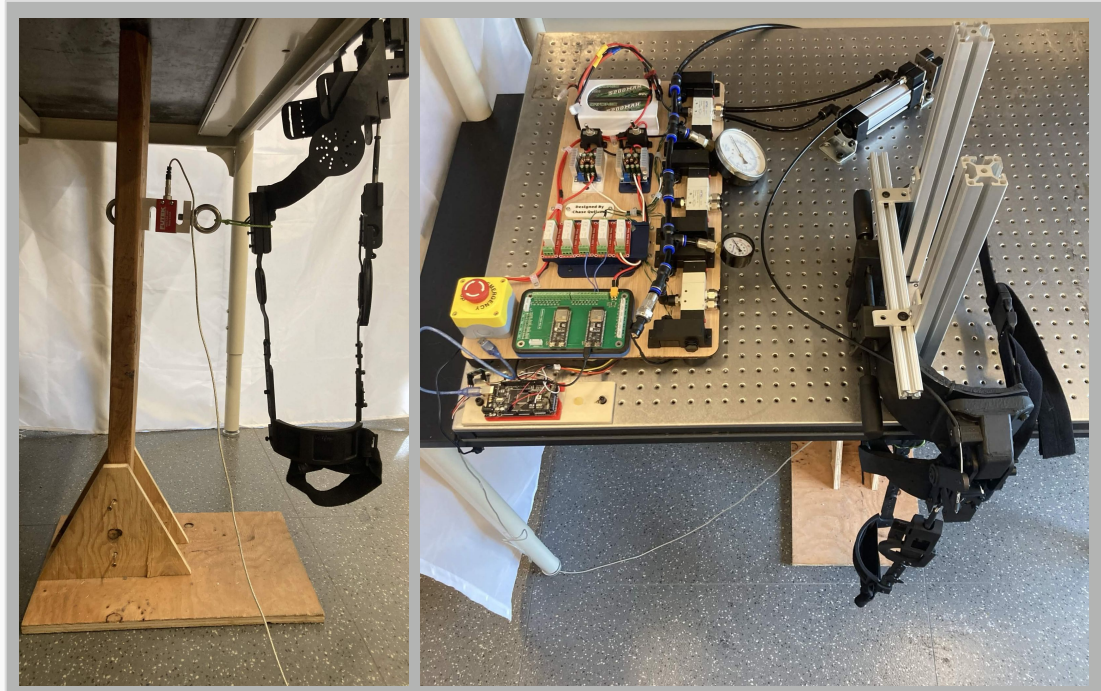
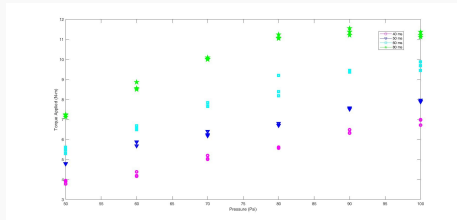
Modular Pneumatic Cylinder Assembly

- Mechanism reverse engineered from a bicycle brake handle
- Cable clips into swiveling keyed hole, and allows for easy removal during testing
- Comprised of $\frac{1}{4}$ " waterjet steel and ABS 3D prints for rigidity



Hip Abduction Load Cell Testing

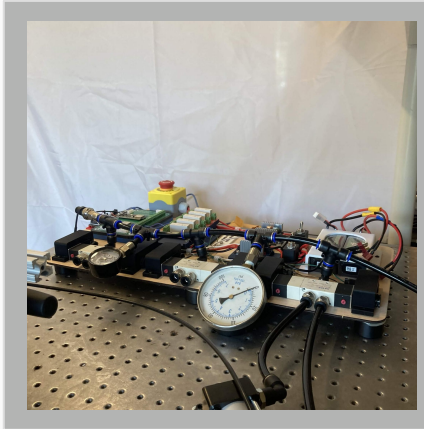
- 96 Total tests have been run at varying delays (40,50,60,80)ms and Psi (50,60,70,80,90,100)
- Data has been deemed viable and trends are still being analyzed



Looking Forward

Onboarding & Documentation

- Onboarding guide
- Finalized test procedures



Testing and Functionality

- Knee abduction DAQ
- Worn exoskeleton DAQ with use of IMU System and MoCap System
- Worn DAQ with Implementation of gait detection algorithm

Hardware/Electrical

- Closed loop control of pneumatic cylinders with string driven potentiometers

Standards and Regulations

Standard	Description
IEC 60601-1	Safe power distribution, grounding, and insulation
CISPR 11, EN 61000	Ensure no interference with other devices
ISO/IEC 14908	Data Exchange
ISO 5725/ ISO 6789	Maintain sensor precision/calibration
ISO 14791	Risk management and safety evaluations
ISO 13850	Emergency stop/disconnect
Pilz 8.3.3, 8.5.2	Installation of pneumatic linear actuators

	2026 Spring Deliverables	
Chassis Improvements	Increased strength and rigidity by machining new metal parts	✓
Load Cell Testing	Performed DAQ on for hip abduction across varying pressures and millisecond delays of actuation	✓
IMU System V2	Designed and built wearable system that performs quaternion data logging at 400Hz sampling rate across 8 IMUs simultaneously	✓
Onboarding & Procedures	Drafted a comprehensive guide for device operation and tuning, complete with detailed sections on various key systems	✓
Test Bench V2	Designed and built test bench that encompasses entire control system, and has been utilized for data collection and testing	✓
Modular Pneumatic Cylinder Assembly	Designed and built mechanism for brake cables to be easily removed from cylinders when needed	✓

Thank you!

And a special thanks to Dr. Trkov, Vaibhavsingh Varma,
and Nick Paglioca our advisors!
